

Time Complexity Examples

1.

```
function addition (a, b){  
    sum = a + b  
    print(sum)  
    print(a)  
    print(b)  
}
```

What is the time complexity here?

Answer: All the lines in the above function are simple statements, which means they take 1 unit of time to execute. Therefore, total time will be 4 units i.e. $O(1)$.

2.

```
function math (a, b){  
    print(a)  
    print(b)  
    sum = a + b  
    print(sum)  
    difference = a - b  
    print(difference)  
}
```

What is the time complexity here?

Answer: All the lines in the above function are simple statements, which means they take 1 unit of time to execute. Therefore, total time will be 6 units i.e. $O(1)$.

3.

```
for (int k = 0; k < 5; k++) {  
    print("a")  
}
```

What is the time complexity here?

Answer: Here, the loop will run for 5 times exactly. Therefore, total time will be 6 units (+1 unit for breaking condition) i.e. $O(1)$

4.

```
for (int k = 0; k < 100; k++) {  
    print("a")  
}
```

What is the time complexity here?

Answer: Here, the loop will run for 100 times exactly.
Therefore, total time will be 101 units (+1 unit for breaking condition) i.e. $O(1)$

5.

```
for (int k = 0; k < n; k++) {  
    print("a")  
}
```

What is the time complexity here?

Answer: Here, the loop will run for n times exactly. Therefore, total time will be n + 1 units (+1 unit for breaking condition) i.e. $O(n)$

Space Complexity Examples

1.

```
function math (a, b){  
  print(a)  
  print(b)  
  sum = a + b  
  print(sum)  
  difference = a - b  
  print(difference)  
}
```

What is the space complexity here?

Answer: The function has two local variables (a and b) and creates two new variables **sum** and **difference**. Therefore, total space required is 4 units i.e. $O(1)$

2.

```
function printArr (n){  
  arr = new array(n)  
  print("a")  
}
```

What is the space complexity here?

Answer: The function creates an array **arr** of size **n**. The array takes up n units of memory space, proportional to its size. Therefore, total space required is n units i.e. $O(n)$.